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Coherent Matter Moonshot

Manufacturing multi-scale materials with ordered liquids at industrial scale

Vision:

Civilization needs a new kind of inexpensive, precision manufacturing. Existing processes tradeoff between costly nanoscale precision, as in semiconductor fabs, and inexpensive bulk shaping with few structural levers, as in plastic film production. Next-generation batteries and membranes for advanced energy systems and critical industrial technologies need affordable precision across scales. Programming multi-scale structures makes their core tradeoffs more tractable: balancing heat, charge and fluid transport in batteries and selectivity against permeability in membranes. The same capability can unlock new materials for advanced sensors and medical applications like spectropolarimetric films, synthetic tendons and corneas.

Multi-scale materials with amazing capabilities are abundantly synthesized in nature by ordering and then solidifying liquid precursors. Gecko feet, moth eyes, and butterfly wings get reversible adhesion, reduced reflection, and dye-free color from structures spanning nano- to macroscopic scales[1,2]. A new platform for programming multiscale structures will let us engineer similar functions by controlling how force, light, charge, and heat move through materials. We call such materials, with designed structures across scales, coherent matter.

The Coherent Matter Moonshot (CMM) is a €35M, three-year coordinated R&D program to build this platform and make coherent matter production inexpensive. Inspired by biological materials, CMM will direct assembly of liquid precursors and preserve their structure through solidification. Its first focus is optical films with structure ordered from nanometers to meters.

Why Now?

We can finally build a shared platform that supports the production of broad classes of multi-scale materials that were previously demonstrated in lab settings due to recent advances unlocking capabilities to design, control and measure coherent matter production industrially:

1. **Design: coherent matter assembly is increasingly predictable**
    - **Barrier:** Due to their weak, multivalent interactions, soft-matter precursors can occupy many metastable states [1, 3, 4]. Reproducibly reaching a given state requires precisely navigating high-dimensional and changing energy landscapes.
    - **What changed**: Predictive models. Self-assembly models can now screen potential precursors and synthesis pathways [5]. Further, AI can now learn from the large datasets that in-line measurement produces[6].

2. **Control: process controls can achieve steer assemblies to targeted designs**
    - **Barrier:** Process controls are entangled. Many factors, such as formulation, flow and drying, jointly set the final architecture [7, 8], so tuning one structural level disturbs the others.
    - **What changed**: Proven methods to direct liquid precursors into desired structures[9] and self-driving labs can optimize multistep processes to learn the optimal trajectory to desired material end-states[6,10].
3. **Measure: new tools enable inexpensive, real-time process measurements**
    - **Barrier:** Diffraction and ellipsometry measure structure precisely in chip production, but they are priced for wafers, not for films worth a few euros per m², and they have not been adapted to processes relevant to CMM[11].
    - **What changed**: Single-shot polarimetry and industrial optical coherence tomography provide real-time structural information affordably [12–14]. When calibrated to other structural measurements, like X-ray scattering[, 15], structure can be inferred rapidly in process conditions.

The challenge is integrating these capabilities into a continuous, controlled production process. Single-material processes exist: cellulose photonic films are made roll-to-roll [16, 17], liquid-crystal modules are in industrial production [18], and roll-to-roll coating lines are mature [19, 20]. But coordinating precursor chemistry, applied fields, solidification and feedback still needs new process development for each new material.

The Programme

Approach. CMM integrates nanoscale self-assembly with programmable control at larger scales, guiding liquid precursors into defined, multiscale structures and preserving them through staged solidification. The platform combines demonstrated technologies to test if a new kind of manufacturing platform is possible.

Scope. The program begins with optical films because optical readout provides the fastest structural feedback and the most immediate markets. Success with optical applications will lead to logical follow-ons and advancements that can extend the platform's proven capabilities.

Use of Funds:

Purpose	**Team**	**Phase 1**	**Phase 2**	**Total**
Core Platform	Core lab team	€10M	€5M	€15M
Control Module Challenge	6 challenge teams	€6M	€6M	€12M
Test Network	3+ partner labs & pilot line	–	€8M	€8M
Total		**€16M**	**€19M**	**€35M**

Phase 1 (Years 1–2): Platform proof and independent verification. A core team (€5M/year) will build the platform. From month 12, competitively selected challenge teams will build control modules. The central research challenge is reproducible control of multi-scale structures with available process controls. Funded work:

- **The Core Prototyping Platform:** A modular benchtop roll-to-roll platform for multi-scale optical films starting with liquid crystals. It includes a minimal set of process controls, an interface specification for the control module challenge (see below), and common measurement modalities. Shared software coordinates the various instruments.
- **Control module challenge** (€12M, years 2–3): six teams funded to build field actuators to a published interface spec (€1M/team for the first year). At month 24, the three teams with the best spec conformance and measured benefit get further funded (€2M/year), and hand-off their modules to the core team by month 30.
- **The multi-scale Materials Trajectory Atlas:**The Atlas connects precursor properties and processing histories to resulting structures, optical responses and manufacturing outcomes. It will identify accessible pathways, phase transitions, kinetic-locking opportunities and operating limits, including unsuccessful trajectories. These maps will define the experimentally supported control space for trajectory planning and feedback.
- **Support Functions:** An external advisory board (EAB) and test network will be recruited (see Appendix).

Phase 2 (Year 3, €19M): Scale-up and handoff The three remaining challenge teams finish their modules, and an €8M tranche runs a Test Network of 3+ labs, grows the Atlas to thousands of trajectories, and retrofits a partner pilot coater (e.g., Fraunhofer FEP [19]) to run Atlas recipes at 1 m²/h. Open specifications v1 follow, and an IMEC-style industrial consortium (Appendix), anchored by the Phase 1 in-kind commitments, commits year-4 funding by month 33 in exchange for early access to the platform, Atlas and IP.

Milestones and gates. Each gate releases the next funding stream; work continues only if its targets are met.

Month	Gates and Milestones
6	Liquid-crystal films produced via roll-to-roll process; challenge call and module interface spec published
12	Gate 1, platform proof of concept: ≥10 m of film at reference spec, in-line optics within 5% of X-ray scattering, first spectropolarimetric film (4-state retarder array, pixels ≤20 μm), and EAB film specs agreed upon.
24	Gate 2, go/no-go criteria (see below) on scaling. Also 0.25 m²/h with order from 10 nm to 10 cm (TRL 5) and optical film with ≥16 spectral × polarization channels
30	Consortium agreement and open specs v1. Pilot roll-to-roll line retrofitted. ≥3 labs

	within 10% site-to-site. Optical film inside a prototype camera
36	Gate 3, handoff: pilot line runs Atlas recipes at 1 m²/h with order from 10 nm to 1 m (TRL 6); techno-economics validated on one EAB spec; consortium takes over

What success looks like. If the Moonshot succeeds, it will produce 1) a fully functional benchtop manufacturing platform deployed across a network of leading research labs, 2) a systematic, open dataset mapping liquid-state control to multi-scale optical function, 3) lab-validated optical materials that solve immediate industrial bottlenecks in computing and sensing, and crucially, 4) the successful ignition of an economic feedback loop where industrial demand for these first-generation materials directly funds the R&D for the second generation.

Falsification and go/no-go criteria. For progression, the core platform in Phase 1 must demonstrate:

- **Controllability**: independently set at least two structural levels (e.g., nanoscale pitch and mesoscale orientation) in one film, with ~10 nm precision in selected features.
- **Reproducibility**: variation in pitch, orientation and optical response below 5% run-to-run and below 10% between the core lab and partner sites.
- **Generality**: two materially distinct precursor classes, liquid crystals [9] and, i.e., cellulose nanocrystals [16, 17], run on the same platform.
- **Economic pull**: 3 EAB members sign in-kind Phase 2 commitments totalling ≥€8M.

Appendix

Visit leronperez.com/BiTSconceptnote for the full appendix and the most recent version of the concept note.

References

[1] E. Amstad and M. J. Harrington, “From vesicles to materials: bioinspired strategies for fabricating hierarchically structured soft matter,” *Philos. Trans. R. Soc. A*, vol. 379, no. 2206, p. 20200338, Aug. 2021, doi: 10.1098/rsta.2020.0338.

[2] U. G. K. Wegst, H. Bai, E. Saiz, A. P. Tomsia, and R. O. Ritchie, “Bioinspired structural materials,” *Nat. Mater.*, vol. 14, no. 1, pp. 23–36, Jan. 2015, doi: 10.1038/nmat4089.

[3] S. Whitelam and R. L. Jack, “The statistical mechanics of dynamic pathways to self-assembly,” *Annu. Rev. Phys. Chem.*, vol. 66, pp. 143–163, 2015.

[4] F. Tantakitti *et al.*, “Energy landscapes and functions of supramolecular systems,” *Nat. Mater.*, vol. 15, no. 4, pp. 469–476, 2016.

[5] M. Dijkstra and E. Luijten, “From predictive modelling to machine learning and reverse engineering of colloidal self-assembly,” *Nat. Mater.*, vol. 20, no. 6, pp. 762–773, Jun. 2021, doi: 10.1038/s41563-021-01014-2.

[6] T. B. Martin, D. Sutherland, A. McDannald, A. G. Kusne, and P. A. Beaucage, “Autonomous small-angle scattering for accelerated soft material formulation optimization,” arXiv:2503.11859, Mar. 2025.

[7] DOE Laboratory Collaboration, “Roll-to-roll advanced materials manufacturing—Early stage R&D Phase 2 and FY2021 final report,” Oak Ridge Natl. Lab., Oak Ridge, TN, USA, Rep. ORNL/TM-2022/2499, 2022.

[8] Creel *et al.*, “Slot-die-coating operability windows for polymer electrolyte membrane fuel cell cathode catalyst layers,” *J. Colloid Interface Sci.*, vol. 610, pp. 474–485, 2022.

[9] R. Telles *et al.*, “Spatially programmed alignment and actuation in printed liquid crystal elastomers,” *Proc. Natl. Acad. Sci.*, vol. 122, no. 3, p. e2414960122, Jan. 2025, doi: 10.1073/pnas.2414960122.

[10] A. A. Volk *et al.*, “AlphaFlow: autonomous discovery and optimization of multi-step chemistry using a self-driven fluidic lab guided by reinforcement learning,” *Nat. Commun.*, vol. 14, no. 1, p. 1403, Mar. 2023, doi: 10.1038/s41467-023-37139-y.

[11] Applied Physics USA, “Semiconductor metrology equipment for sale: buyer’s guide.” Accessed: Oct. 27, 2026. [Online]. Available: https://appliedphysicsusa.com/semiconductor-metrology-equipment-for-sale-buyers-guide/

[12] A. Zaidi *et al.*, “Metasurface-enabled single-shot and complete Mueller matrix imaging,” *Nat. Photonics*, vol. 18, no. 7, pp. 704–712, Jul. 2024, doi: 10.1038/s41566-024-01426-x.

[13] S. Sacher *et al.*, “Shedding light on coatings: Real-time monitoring of coating quality

at industrial scale," *Int. J. Pharm.*, vol. 566, pp. 57–66, Jul. 2019, doi: 10.1016/j.ijpharm.2019.05.048.

[14] N. S. Kahatapitiya *et al.*, "Optical coherence tomography for high-precision industrial inspection in Industry 4.0: Advances, challenges, and future trends," *Laser Photonics Rev.*, vol. 20, no. 8, p. e02290, 2026, doi: 10.1002/lpor.202502290.

[15] DESY Photon Science, "P03: The micro- and nanofocus X-ray scattering beamline at PETRA III." [Online].

[16] H.-L. Liang *et al.*, "Roll-to-roll fabrication of touch-responsive cellulose photonic laminates," *Nat. Commun.*, vol. 9, no. 1, p. 4632, Nov. 2018, doi: 10.1038/s41467-018-07048-6.

[17] B. E. Droguet *et al.*, "Large-scale fabrication of structurally coloured cellulose nanocrystal films and effect pigments," *Nat. Mater.*, vol. 21, no. 3, pp. 352–358, Mar. 2022, doi: 10.1038/s41563-021-01135-8.

[18] Photonics.com, "Merck opens new liquid crystal module production facility." [Online].

[19] Fraunhofer FEP, "Roll-to-roll pilot lines," equipment flyer. [Online].

[20] Perovskite-Info, "Coatema coating machinery." [Online].